DOMS Page No.
Date / /

ASSIGNMENT : 4

Problem Statement: Write a C program to calculate the salary of an employee given his basic pay. Calculate gross salary of employee. Let HRA be 10% of basic pay & TA be 5% of basic pay. Let employees pay professional tax as 2% of total salary. Calculate net salary payable after deductions.

Objectives:
1. To understand the importance of flowchart for any programming model.
2. To learn simple flowchart symbols & arrows to define relationships
3. To understand & develop visual representations of flow of data.

Theory:
1. Define Algorithm

A. An algorithm is a finite set of well-defined, step-by-step instructions used to solve ~~to~~ a problem or perform a task. It takes input, processes it through logical steps, & produces an output.

Key characteristics:
a) Input: Accepts zero or more inputs.
b) Output: Produces atleast one output.
c) Definiteness: Each step is clear & unambiguous.
d) Finiteness: Terminates after finite number of steps.
e) Effectiveness: Each step is simple & executable.

DOMS Page No.
Date / /

2 What are the different characteristics of an algorithm.

A. a) Input: An Algorithm should accept zero or more inputs from the user or a given source.
b) Output: It must produce at least one output as result of processing the input.
c) Definiteness: Each step of the algorithm should be clear, precise and unambiguous.
d) Finiteness: The algorithm must terminate after a finite no. of steps.

Platform: Linux

Input: Basic Pay (entered by user)

Output: Display
Basic Pay
HRA
TA
Gross Salary
Professional Tax
Net Salary Payable

Conclusion: Thus, we learned to write algorithms & to draw its representation in terms of flowchart.

FAQs:
1. Describe various symbols & their use in flowchart
A. 1. Oval: To start & STOP.
2. Parallelogram: For Input & Output purpose.

3. Rectangle: Represents processing steps such as calculations, assignments or instructions.
4. Diamond: Represents a decision or condition that results in branching.
5. Flow line: Shows directions of flow & connects different symbols.

2. Difference between Algorithm & Flowchart.

Algorithm	Flowchart
a. An algorithm is a step-by-step procedure to solve a problem.	a. Flowchart is graphical representation of steps of an algorithm.
b. Written in simple English or pseudocode.	b. Uses a standard symbols like oval, rectangle, arrows etc.
c. Easy & quick to write & modify	c. Easy to understand & visualize the program lo
d. Suitable for complex logic description.	d. Best suited for visual explanation of logi
e. Focuses on logic & steps	e. Focuses on flow of control & data.